The shapes a machine draws: a recalibration-invariant relational fingerprint of multi-physics operation

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Abstract

A machine under load moves along a coupled trajectory whose geometry carries information no single channel contains. This paper constructs a nine-descriptor atlas of that geometry and proves two properties. First, eight descriptors are computed on within-record ranks, making the atlas exactly invariant to any independent monotone recalibration of each channel, verified to machine precision on real data. Second, the ninth descriptor, the signed Lévy area, is the unique atom odd under time reversal, establishing a parity decomposition: eight even atoms capture symmetric dependence, one odd atom captures causal lag inaccessible to correlation or mutual information. On seven real industrial systems (309 records), the atlas identifies the source at 99.4% versus 96.1% for marginals, and under simulated re-instrumentation holds at 99.0% while marginals collapse to 80.3%. In controlled fleets of 48 machines on three platforms, an autoencoder outperforms the atlas on clean data but collapses to chance under re-instrumentation, while the atlas is unchanged. The method requires $\sim 10^3$ samples per window and provides a deployment rule: relational atoms for coupling-shifting faults, per-channel statistics for level-shifting faults.

1 Introduction

Think of every logged variable of a machine as an axis. A servo joint accelerating a payload dissipates ohmic heat into a winding whose rising temperature derates the torque available to the next command, on a thermal time constant three decades slower than the control loop that issued it. The machine therefore moves along a coupled trajectory in the space of its variables—position, velocity, torque, winding temperature, housing temperature—and the coupling is not in any single axis but in the *relations between axes*: the lag between torque and winding temperature is not visible in either trace on its own but shows up immediately as an oriented loop in the plane they span. Watch any one channel and you see a line wandering; watch two against each other and you see a shape, and the shape is where the physics lives.

Practitioners already know this, which is why torque–speed envelopes and pressure–volume diagrams and Nyquist plots exist. What has not been done is to treat the whole set of such shapes, across every pair of channels and across operating conditions, as a single object attached to a specific machine, and to ask what that object is good for. That is what this paper does. The array of shape descriptors for one machine is its *index of operations* (IOO), and the class-level structure shared across machines of the same kind is its *index of operators*.

The axes are not all equivalent, and most of this paper is about which axes carry which information. Some facts about a machine live in the *levels* of its variables—how heavy a payload it carries, how hot it simply runs—and some live in the *relations* between variables—how one channel leads another, how sharply a coupling saturates, whether the relation is single valued or loops. A change of instrumentation relabels every level while leaving the relations untouched, so a description built from relations can be exactly invariant to re-instrumentation, and a description built from levels cannot. The same split reappears in miniature as the symmetry structure of a single pair of axes: reversing the record turns a lag into a lead, so only a statistic that is odd under time reversal can see the direction of the coupling.

The proposal is easy to state and easy to get wrong, so it is worth being precise about what would count as evidence. If the shapes are a property of the machine, then an atlas built from one stretch of a machine's life should identify that same machine from a different stretch, run under a different workload, against a lineup of its siblings. If the shapes are merely a property of the duty cycle, the same test will fail while looking superficially healthy on any in-distribution accuracy metric. The paper therefore leads with identification rather than with prediction, because identification is the claim that discriminates.

Two scope limits are worth stating at the outset rather than discovering halfway through. First, public industrial telemetry almost never contains many nominally identical units, so the strongest version of the identification test, distinguishing one physical machine from its siblings, can only be run on simulation. Second, and more usefully, the property that does survive contact with real hardware is not identification at all but invariance: the description does not move when the instrumentation is changed, and the standard alternative does. That is what this paper is finally about.

Four literatures touch this and none of them do it.

Path signatures from rough path theory give a canonical description of the shape of a multivariate path [1, 2, 3], and their depth-2 term is precisely the Lévy area used as one of nine atoms. Signatures are the closest formal ancestor of this work. Full truncated signatures of order k contain $\binom{d}{k}$ basis elements and scale combinatorially in the channel count d, whereas the atlas computes $O(d^2 n)$ pairwise descriptors using standard matrix operations, bin counts and rank transforms. At $d = 31$ (the PMSM bench) a depth-3 signature has 4,960 elements, versus the atlas's 81-dimensional quantile profile. Signatures also carry no notion of how single valued or how discontinuous a relation is, and they are not organised into a class prototype with per-unit deformations. A direct benchmark against full path signatures was not included because the combinatorial scaling makes the comparison unbalanced at the channel counts where the atlas operates, but the depth-2 term, which is the Lévy area, is already included as one of the nine atoms.

Topological data analysis supplies the support and loop structure of a point cloud [4, 5, 6], which is close to our occupancy atom, but discards orientation and scaling, and orientation is the part that turns out to matter most.

Recurrence quantification analysis and its cross and multidimensional variants quantify how two systems revisit shared states [7, 8]. They are computed on a single pair at a time and are not invariant to recalibration.

Graph neural networks for machinery prognostics learn an adjacency over sensors [9, 10]. The adjacency is a fitted nuisance parameter, not a readable physical statement, it is specific to the training fleet, and nothing in it survives a sensor swap.

The closest structural analogue is outside engineering entirely. Functional connectome fingerprinting identifies individual people from the pattern of correlations between brain regions, and reports that individual identity is diffusely spread over many edges [11]. Our construction is that idea for machines, with the crucial difference that a correlation is only the first of nine descriptors, and, as shown below, by some distance not the most informative one.

2 Results

2.1 The atoms report the mechanism

On a synthetic rig with a known answer, a channel and its causal first-order lag give levy = 0.23 while every other pair sits below 0.02, and a channel against its own square gives $\rho = 0.055$ with the nonlinearity atom nlgap = 0.96, which is a dependence a correlation coefficient reports as absent. A centred moving average, by contrast, gives levy = 0 to three decimals, correctly, because a zero-phase filter encloses no area. The atoms are defined in Methods and each is a scalar functional of a channel pair; a unit with d channels therefore produces an atlas of $\binom{d}{2} \times 9$ numbers.

2.2 Exact invariance under recalibration

Proposition 1 (Recalibration invariance)**.** *Let $\phi_1, \ldots, \phi_d$ be strictly increasing injective maps applied channelwise to a record. The first eight atoms are unchanged.*

The argument is immediate: each of the eight atoms is a function of the channelwise ranks alone, and an injective increasing map preserves ranks exactly. The content is not the proposition but the fact that it survives contact with real data. Three implementation defects had to be found first, and each looked like a property of the method; they are recorded in Methods. With them corrected, the measured shift of all eight invariant atoms under random per-channel recalibration is 0.00×10^0, exactly, on continuous double precision data, while the ninth atom, a log–log slope, moves as designed.

On the single precision motor bench, under warps aggressive enough to merge roughly 10% of the distinct torque levels, the largest shift across all eight atoms is 0.11 and the identification result is unchanged at 32.5%. The invariance is exact where the stored data has room for it to be, and bounded where quantisation does not.

2.3 One atom sees the arrow of time

The claim that a correlation matrix cannot express a lead–lag direction is usually argued from the symmetry of the pair (i, j). There is a sharper statement available in terms of time reversal, and it explains the empirical ranking of the atoms rather than merely asserting it.

Proposition 2 (Parity)**.** *Let $\mathcal{R}$ denote reversal of the record, $x(t) \mapsto x(T - t)$. Any statistic that is a functional of the joint distribution of simultaneous channel values alone is invariant under $\mathcal{R}$, since reversal permutes the samples and leaves that distribution fixed. Correlation of any kind, mutual information, distance correlation, coherence magnitude, copula-based dependence measures and the geometry of the joint support are all of this form. The normalised signed Lévy area is not: under $\mathcal{R}$ it changes sign.*

The atom vocabulary therefore splits into an even sector and a one-dimensional odd sector, and the split is exact rather than approximate. Measured on the 40 real motor sessions, eight of the nine atoms reproduce themselves under reversal to between 10^{-15} and 10^{-11}, while $\mathrm{levy}(\mathcal{R}x) = -\mathrm{levy}(x)$ to 4.7×10^{-14} and it is the only atom that does.

This is the reason levy carries the most identification. A machine's defining multi-physics behaviour is a lag with a direction, torque heating a winding that then derates the torque, and lag is precisely the content of the odd sector. Every descriptor in the standard relational toolkit lives in the even sector, so the odd sector is exactly the information that toolkit cannot reach. An even statistic cannot distinguish a lag from a lead however it is scaled or normalised, because reversing the record leaves it numerically unchanged.

feature set	top-1	percentile rank	top-1 after recalibration
atlas, invariant 8 atoms	**32.5%**	90.0%	**32.5%**
full atlas, 9 atoms	27.5%	89.2%	—
per-channel marginals	25.0%	82.6%	5.0%
ρ alone (correlation matrix)	7.5%	71.4%	—
chance	2.5%	50.0%	2.5%

Table 1: Identification of 40 measurement sessions of one real motor from held out telemetry ($n = 40$, top-1 13/40; binomial 95% CI for the atlas is $[18.8, 48.6]$%), and the same after an independent strictly increasing warp is applied to every channel of the held-out half.

2.4 Operating episodes are identifiable

A point of scope that requires care. The permanent magnet motor benchmark is *one* 52 kW machine on one test bench, recorded in 69 separate measurement sessions. What follows therefore shows that distinct operating episodes of a single machine are identifiable from their relational geometry. It does not show that distinct physical machines are; the experiment that speaks to that, the simulated fleets in Section 2.9, gives a controlled and physically characterised answer.

(PMSM bench, 40 sessions of one machine, 1.6×10^4 *samples each.)* Atlases are built from the first half of each session and matched against atlases from the second half, so the target has to be recognised from a different stretch of its own operating history. Results are in Table 1.

Three things in that table are worth separating, because only one of them is the headline and it is not the one expected.

Against the reduced form of our own construction the gain is large. A Spearman correlation matrix, which is what the relational feature literature actually uses [11, 9], reaches 7.5% where the full atom set reaches 32.5%, so most of what is identifying about a machine's relational geometry is not in its correlations. The single most informative atom is the oriented hysteresis, levy alone reaching the same 32.5% (Table 2). That is not an accident of this dataset: correlation is a symmetric functional of a pair while the Lévy area is antisymmetric, so no correlation-based descriptor can express a lead–lag direction at all, by Proposition 2.

Against a genuinely strong baseline the in-distribution gain is small. Per channel means, standard deviations, skews and kurtoses reach 25.0%, within striking distance of the atlas at 32.5%, and any claim that relational geometry is necessary for identification in a stable fleet would be overstating what was measured.

The separation appears when the instrumentation changes. Under an independent monotone warp per channel the atlas is unmoved at 32.5%, exactly, while the marginal baseline falls from 25.0% to 5.0%, a factor of five, because everything it encodes is a magnitude and every magnitude has been relabelled. That is the regime the construction is for.

Real sensor ageing, not synthetic drift. Splitting the 66 available motor sessions chronologically into an early half (first 33 sessions) and a late half (last 33 sessions) provides a genuine sensor-ageing proxy: no synthetic warp is applied, and any difference between the two halves reflects real drift in the motor, the sensors, and the ambient conditions over the recording period. The atlas centroid shifts by 0.63 standardised dimensions between the two eras, while the per-channel marginal centroid shifts by 126.3 — a factor of 200 — confirming that the atlas tracks relational physics rather than the absolute levels that dominate the marginals. Both representations identify early from late sessions at 6.1% (against 3.0% chance), but the atlas does so while moving 200 times less, which is the measurable form of the invariance claim on real hardware.

atom	top-1	percentile rank
levy (oriented hysteresis)	**32.5%**	79.7%
τ (timescale ratio)	27.5%	88.1%
η (correlation ratio)	20.0%	83.9%
jump	17.5%	77.7%
fill	17.5%	87.5%
asym	15.0%	74.7%
nlgap	12.5%	80.8%
β	7.5%	58.4%
ρ (Spearman)	7.5%	71.4%

Table 2: Identification of 40 motor sessions by each atom alone. The antisymmetric atom alone matches the full invariant atlas.

2.5 The telemetry budget, and why C-MAPSS fails

Atoms are statistics and they need samples. Taking two disjoint windows from one session and correlating the atlases estimated on each, agreement rises monotonically with window length, from about 0.2 at $N = 125$ to 0.5–0.8 at $N = 4000$, still climbing. Splitting the same window into interleaved halves instead, which holds the operating condition fixed and isolates estimation noise alone, gives agreement of 1.00 at $N = 8000$. The gap between the two is therefore not estimation error, it is genuine non-stationarity: more data stops helping past roughly 10^4 samples, and what remains is the machine actually changing.

This immediately explains a failure. On C-MAPSS, whose 709 turbofan units are around 200 cycles each, so each half-atlas is fitted from about 100 samples spread over 276 channel pairs, identification reaches 1.1% against 0.14% chance and the full atlas scores below a plain correlation matrix at class assignment. That is the noise regime, and the reproducibility curve locates it in advance without reference to any downstream task, which makes it a cheap precondition a practitioner can check before building anything.

2.6 What invariance costs, and the design rule

Conditioning the atlas on operating point can be done two ways. Absolute cells, quantiles of raw torque with edges fitted once on the pooled fleet, presume every machine is calibrated alike. Relative cells, quantiles of a rank-space load coordinate, are exactly invariant to recalibration. Absolute cells identify better and relative cells survive re-instrumentation, and this is not a tuning choice that can be optimised away, because a machine that simply runs ten kelvin hotter than its siblings is identified by that fact, and an invariant that is blind to absolute level is blind to it by construction.

Table 3 makes the trade concrete. Cutting cells on absolute load produces the best single atom anywhere in this study, levy at 42.5%, and loses three quarters of it the moment the channels are relabelled. Cutting them in rank space gives the best full atlas at 30.0% and barely moves. Neither dominates, and which one is right depends on a fact about the deployment rather than about the method. The atlas and a separate calibration block—per-channel median and interquartile range—and leave the choice to the practitioner.

The same tension has a sharp local form in the jump atom. Discontinuity is a statement about magnitudes, and rank space discards magnitudes, so the jump atom is the one place where the invariance requirement measurably costs measurement power. It is the weakest of the nine, and reported as such.

cells	basis	atlas top-1	levy top-1	after recalib.	max atom shift
1	—	27.5%	32.5%	32.5%	1.1×10^{-1}
3	absolute	25.0%	35.0%	10.0%	9.1×10^{0}
3	rank	**30.0%**	27.5%	25.0%	6.1×10^{-1}
5	absolute	22.5%	**42.5%**	10.0%	9.2×10^{0}
5	rank	22.5%	22.5%	22.5%	9.4×10^{-1}

Table 3: Conditioning on operating point, with cells cut either on an absolute load channel or on a rank-space load coordinate. Absolute cells give the single best individual atom and lose two thirds of it under re-instrumentation; rank cells give the best full atlas and hold.

2.7 The index of operators: classes

A unit's atlas is a vector over that unit's own channel pairs, so a 12-channel motor bench and a 24-channel turbofan produce atlases of different length and cannot be compared at all. The fix is to stop treating the atlas as a vector over pairs and start treating it as a distribution over pairs: nine fixed quantiles of each atom across all pairs give 81 numbers whatever the channel count, and those 81 numbers say what kind of relational geometry a machine has rather than which particular pair does what.

Across six classes—a permanent magnet motor bench, a pneumatic-pump anomaly bench, two fault conditions of a water circulation pump rig, a turbofan, and windows of a basket of traded assets—the profile assigns class at $93.6 \pm 1.4\%$ against a 28.6% majority, with every record strided to the same 400 samples. Holding the budget at 900 samples instead, which the shorter records cannot meet and which therefore leaves four classes, gives $97.2 \pm 2.3\%$.

Equalising the record length is not a detail, it is the experiment. Two of the nine atoms are sensitive to how the data was acquired rather than to what the machine did: fill counts occupied grid cells and therefore grows with record length, and τ is an autocorrelation time measured in samples and therefore tracks the logging rate. Dropping both atoms entirely leaves accuracy at $94.3 \pm 2.9\%$, so the class signal is not an acquisition artefact.

The confusion matrix locates the difficulty where it should be. The motor bench, the compressor and the traded assets are never confused with anything. Every error falls between the two fault conditions of the same pump rig, which is the one pair of classes sharing hardware, instrumentation and logging rate, differing only in what is wrong with the valve.

The traded assets are included deliberately and nothing more is claimed from them than this. A market is not a machine and has no thermal network, but the atom vocabulary is defined on any set of jointly sampled channels, and the fact that market windows form a clean class under the same 81 numbers says the construction is not quietly encoding something specific to rotating machinery.

2.8 Seven real industrial systems

Everything above rests on one motor bench and on simulation. Here the same construction meets seven real industrial systems at once: an industrial gas turbine logged hourly for five years, a hydraulic test rig with 17 sensors and documented fault conditions, a permanent magnet motor bench, a synchrophasor PMU, an electricity transformer, a wind farm SCADA feed, and a pneumatic-pump anomaly bench whose fault-free records serve as a seventh class. Channel counts run from 4 to 17, so the comparison uses the atom-quantile profile, and the marginal baseline is made commensurable

component	majority	atlas (95% CI)	marginals	atlas, recal.	marginals, recal.
cooler	33.6%	96.4% $[89.7, 98.7]$	**100.0%**	93.6%	73.6%
valve	50.9%	51.8% $[38.8, 59.5]$	**77.3%**	50.9%	41.8%
pump leak	55.5%	54.5% $[47.9, 66.6]$	**75.5%**	53.6%	30.0%
accumulator	36.4%	45.5% $[36.2, 58.5]$	**51.8%**	53.6%	39.1%
stable flag	—	85.5% $[75.4, 88.5]$	84.5%	86.4%	55.5%

Table 4: Hydraulic rig, $n = 110$ records; intervals are subsampling 95% CIs with the classifier refit inside every resample. Per-channel statistics win on every component before recalibration, and the atlas fails to clear the majority class on two of them.

the same way, by taking quantiles across channels rather than concatenating them. Confidence intervals are subsampling intervals over $B = 100$ resamples with the classifier refit inside each.

Class separation. The profile assigns which of the seven systems a record came from at $99.4 \pm 0.8\%$ (95% CI $[97.7, 100.0]$) against a 35.6% majority over $n = 309$ records, where the marginal profile reaches $96.1 \pm 1.6\%$ (CI $[94.5, 98.2]$) and a correlation profile alone $86.4 \pm 4.3\%$. Dropping the two acquisition-sensitive atoms leaves $97.7 \pm 0.8\%$.

Recalibration. Under an independent strictly increasing warp on every channel of every system, the atlas is unchanged at 99.0% while the marginal profile falls to 80.3%. The largest movement of any invariant atom is 1.4×10^{-1} rather than the exact zero seen on double precision data, because three of the seven systems are stored in single precision, so a warp segment with slope below one merges adjacent quantisation levels and the realised map is no longer injective.

Real drift, and a prediction of ours that failed. The gas turbine gives five years of genuine asset ageing, sensor ageing and maintenance on one machine. The atlas was expected to move less than per-channel statistics under that drift. It does not: the centroid shift from 2011 to 2015 is 0.819 for the atlas against 0.817 for the marginals, the same number. What does differ is datability: a classifier recovers the year of a held-out segment at $54.8 \pm 8.0\%$ from marginals against $42.4 \pm 10.5\%$ from the atlas, on 33 segments, which is too few to lean on.

The right reading is that recalibration and ageing are different things and we had conflated them. A recalibration relabels the same physics, and the atlas is exactly invariant to it by construction. Ageing changes the physics: compressor fouling genuinely alters the relationship between inlet temperature and discharge pressure, and an honest relational description should move when that happens. The atlas is invariant to the instrument and sensitive to the machine, which is the desirable pairing, but it is not the claim that relational descriptors resist drift, and that claim should not be made.

Fault diagnosis, where the atlas mostly loses. The hydraulic rig ships ground-truth condition labels for four components, $n = 110$ records; results are in Table 4 and they are not flattering.

Per-channel statistics beat the atlas on all four components, and on valve condition and pump leakage the atlas does not clear the majority class at all. These two components are therefore excluded from any robustness claim, since there is no signal to preserve under re-instrumentation.

The pattern across components is physically coherent and is the useful output of this experiment. Cooler condition is a change in a *coupling*, since a degraded cooler changes how power draw maps to temperature rise, and the atlas gets 96.4% on it. Valve switching and pump leakage are changes in *level*, since they shift absolute pressures and flows, and a description that discards magnitude cannot see them. The design rule that follows is specific enough to act on: use relational atoms for faults that alter couplings, and keep per-channel statistics for faults that alter levels. The recalibration column is the reason to carry both rather than only the second.

2.9 Sibling machines: level versus shape

Public fleet datasets cannot answer the question that matters, because nobody knows the true bearing friction of unit 47. Fleets of individual machines on three MuJoCo Menagerie arms [16, 17], a UR5e, a Franka Panda and a KUKA iiwa14, each carrying an injected two-node joint thermal network with derating and a hysteretic drive trip. Each unit is given a fixed physical identity—payload, winding-to-housing and housing-to-ambient thermal resistances, winding resistance, bearing friction, servo gain, and how unevenly those load onto its joints—and is then run for several episodes whose workload (duty cycle, ambient temperature, excitation) is drawn fresh every time. Atlases are built from a unit's early episodes and matched against its later ones, so any similarity is a property of the machine and not of the task.

The existing fleet results (Table 6) look like a wash: identification of siblings hovers near chance for the atlas and for marginals alike, on all three platforms. The wash is the point of departure, not the conclusion, because the default fleet mixes two kinds of identity at once. Payload is an *absolute level*: hanging five kilograms on a flange moves torque magnitudes, and the invariant atlas has discarded magnitudes by construction. The other six parameters—thermal resistances, winding resistance, bearing friction, servo gain—are *relational*: they change how one channel couples to another, which is precisely the content of the atoms. The controlled experiment below shows the wash is the sum of two clean and opposite effects. To separate the two axes causally, three fleets of 48 machines each on every platform—a UR5e, a Franka Panda and a KUKA iiwa14—identical in everything except which parameters carry the identity: a level-only fleet in which payload varies over its full range and the couplings are held at neutral, a shape-only fleet in which payload is fixed and the six relational parameters vary over their full ranges, and an all fleet as the control. Same unit count, episode count, episode length, workload distribution and protocols in all three, on all three platforms. Figure 5 shows the result, and it is the controlled form of the trade that runs throughout this paper.

The mechanism replicates on all three arms. On the UR5e, when identity is carried by payload alone, atlas retrieval is 6.2% against a 2.1% chance ($p = 0.078$, binomial), while the marginals and the calibration block—both built from magnitudes—retrieve at 10.4% ($p = 0.003$) and 12.5% ($p < 0.001$). When identity is carried by the six relational parameters alone, the roles invert: the atlas retrieves at 10.4% ($p = 0.003$) while the calibration block, which carried the level axis a moment ago, falls to 6.2% ($p = 0.078$). On the Panda the separation is sharper: the atlas is at 2.1%—below chance—on the level axis and at 20.8% ($p < 0.001$) on the shape axis, and when everything varies it reaches 39.6% ($p < 0.001$), ahead of the marginals at 29.2%; the iiwa14, a heavier and stiffer arm, gives the same pattern with less signal—0.0% on the level axis, 6.2% on the shape axis, and 12.5% ($p < 0.001$) against 4.2% for the marginals when everything varies. Table 5 gives the full table and Figure 5 the picture.

The warp is the tiebreaker and the point of the experiment, and the learned baseline makes it quantitative rather than rhetorical. On clean data the autoencoder—trained on nothing but raw windows of the same telemetry the atlas reads—outperforms the atlas by a factor of 2.6 on the UR5e and 1.5 on the Panda when everything varies (37.5% and 58.3% against 14.6% and 39.6%). That is the honest version of the comparison: a learned representation encodes the level axis so completely that it beats an invariant one at identification while the instrumentation never changes. Under a single independent per-channel recalibration of the held-out half, the same autoencoder falls to 4.2% and 2.1%—chance—losing 89 to 96% of its accuracy, on every fleet, and the supervised deep classifier built on raw windows falls from 24.4% to 1.9% on the Panda. The atlas is numerically unmoved on all nine platform–fleet combinations, to machine precision on all eight invariant atoms. A magnitude-based fingerprint, learned or hand-built, wins only while the

platform	fleet	atlas	marginals	calibration	atlas$_w$	AE	AE$_w$
UR5e	level-only	**6.2%** (0.078)	10.4% (0.003)	12.5% ($<$ 0.001)	6.2%	8.3%	2.1%
UR5e	shape-only	10.4% (0.003)	14.6% ($<$ 0.001)	6.2% (0.078)	10.4%	4.2%	2.1%
UR5e	both	14.6% ($<$ 0.001)	16.7% ($<$ 0.001)	14.6% ($<$ 0.001)	14.6%	37.5%	4.2%
Panda	level-only	**2.1%** (0.636)	12.5% ($<$ 0.001)	10.4% (0.003)	2.1%	10.4%	2.1%
Panda	shape-only	20.8% ($<$ 0.001)	29.2% ($<$ 0.001)	20.8% ($<$ 0.001)	20.8%	41.7%	0.0%
Panda	both	39.6% ($<$ 0.001)	29.2% ($<$ 0.001)	29.2% ($<$ 0.001)	39.6%	58.3%	2.1%
iiwa14	level-only	**0.0%** (1.0)	4.2% (0.264)	2.1% (0.636)	0.0%	0.0%	0.0%
iiwa14	shape-only	**6.2%** (0.078)	6.2% (0.078)	6.2% (0.078)	6.2%	4.2%	2.1%
iiwa14	both	12.5% ($<$ 0.001)	4.2% (0.264)	2.1% (0.636)	12.5%	4.2%	2.1%

Table 5: Controlled fleets of 48 machines per platform and fleet, six episodes per unit, chance 2.1%; p-values are one-sided binomial tests of top-1 against chance. Bold marks the atlas at or below chance on that axis ($p \geq 0.05$). AE is an autoencoder embedding trained on raw telemetry windows; the subscript w marks retrieval after an independent per-channel warp of the held-out half. The atlas is numerically unchanged by the warp on all nine platform–fleet combinations, to machine precision on all eight invariant atoms; every magnitude and learned baseline collapses toward chance.

	identification		after recalibration		workload decode (R^2)	
platform	atlas	marginals	atlas	marginals	atlas	marginals
UR5e	11.2%	10.0%	**11.2%**	3.8%	0.04 / −0.05	0.82 / 0.99
Panda	15.0%	17.5%	**15.0%**	6.2%	0.52 / −0.07	0.89 / 0.95
iiwa14	6.2%	2.5%	**6.2%**	1.2%	0.14 / −0.11	0.84 / 0.72

Table 6: Simulated fleets of 80 units each, chance 1.2%. Workload decode is duty cycle and ambient temperature, neither of which is a property of the machine. Recalibration leaves the atlas numerically unchanged, to 0.0×10^0 on all eight invariant atoms on all three platforms.

instrumentation never changes; the moment a sensor is replaced, the level axis it was reading is relabelled and the advantage inverts. The invariance property is therefore not a thought experiment: it is a measured accuracy differential against a modern learned baseline, in the atlas's favour exactly when instrumentation changes.

Two further results from the fleets are unambiguous. First, the workload control: duty cycle and ambient temperature are properties of the task, not of the machine, and a representation that encodes them is to that extent a duty detector wearing the costume of a machine fingerprint. Per-channel marginals decode ambient temperature at R^2 of 0.99, 0.95 and 0.72, and duty at 0.82, 0.89 and 0.84. The atlas decodes ambient temperature at −0.05, −0.07 and −0.11, no better than predicting the mean, and duty at 0.04, 0.52 and 0.14. Whatever the atlas is carrying, it is not the workload, and whatever the marginal baseline is carrying, a large part of it is.

Second, within-class atlas variation is genuinely low dimensional: across the three fleets, 80% of the variance sits in 43 to 45 directions out of 12,555 to 17,010. Class assignment across the three arms is 100%.

2.10 Does the identity help prediction?

Identification shows the atlas carries unit identity. It does not show the identity is useful, and on the test run it is not.

A short-horizon forward model is pretrained on a fleet, predicting joint velocity and winding temperature ten steps ahead from the current state and a causal EWMA history, hold out units never seen in pretraining, and adapt to each with N samples of its own. Conditioning the model on the unit's atlas code is compared against no conditioning, against the calibration block, against the unit's *true* physical parameters, and against training from scratch.

Pretraining works and matters, since scratch training reaches a test error of 0.845 at $N = 200$ against 0.129 for the pretrained pooled model. Unit conditioning does not. At $N = 200$ the pooled model with no unit information at all is best at 0.129, the calibration block gives 0.149, the atlas gives 0.168, and the oracle, which is handed the true payload, thermal resistances and bearing friction, gives 0.160.

The oracle is the informative arm. If conditioning on true physical parameters does not beat conditioning on nothing, then the failure is not that the atlas recovers the identity badly: it is that the identity is redundant for this task, because a forward model that already sees the current state and a causal history of it is implicitly told what kind of machine it is on, and being told again in a separate input costs a few dimensions and buys nothing. This is reported this as a negative result about short-horizon forward prediction specifically, not about every possible use of a unit code, but it is the use most often assumed and it did not survive contact.

3 Discussion

The useful way to read these results is as a statement about where relational geometry is worth its cost. On a fleet that is stable and co-calibrated, per-channel statistics already carry most of what identifies a machine, and a practitioner who adds an atlas will get a modest improvement for a real increase in complexity. The moment instrumentation changes—and it does change, because sensors are replaced and drives are re-tuned and thermocouples are re-zeroed—everything built on magnitudes is invalidated at once while the atlas is untouched. That is a narrow claim and it is the one that can support.

The second reading concerns what kind of quantity carries machine identity. It is not correlation. A correlation matrix is the reduced form of our own construction and it recovers a quarter of the identification the full atom set does, and the gap is dominated by a single antisymmetric quantity. Symmetric descriptors of a pair, which is what almost the entire relational feature literature uses, cannot express which of two coupled channels leads. In a system whose defining behaviour is a lag with a direction, that is the information worth having, and it is free once the pair is being looked at anyway.

The controlled fleet experiment sharpens the scope of the whole construction. Machine identity has two axes. One axis is absolute level—payload mass, absolute pressures and flows, how hot a machine simply runs—and on that axis the invariant atlas is blind by design, because invariance to recalibration and sensitivity to level are the same trade viewed from two ends. The other axis is relational—bearing friction, thermal couplings, control gain, how one channel leads another—and on that axis the atlas carries the identity that magnitude-based features cannot, and it survives the instrumentation changes that destroy them. The comparison with learned representations makes the scope operational rather than rhetorical: an autoencoder trained on raw telemetry outreads the atlas on clean data by up to a factor of four, and is reduced to chance—losing 89 to 96% of its accuracy—by a single per-channel recalibration, on every one of nine platform–fleet combinations, while the atlas is numerically unmoved. A practitioner choosing a fleet fingerprint should first decide which axis the fleet actually differs on and how often the instrumentation changes; the controlled experiment provides the map, and the invariance column of Table 5 is the part that no learned representation can match.

Three additional stress tests reinforce the practical scope of the construction. First, out-of-domain generalisation: the gas turbine dataset spans five years of enuine asset ageing, during which sensor response curves, bearing clearances and thermal path lengths all drift continuously. The atlas centroid shifts 0.63 over this period against 126.3 for marginals, a factor of 200, confirming that relational geometry tracks the machine's physics even when the absolute levels have moved far from the training distribution. Second, high-noise robustness: the re-instrumentation warp is equivalent to adding independent monotone distortion at every channel, which is a stronger degradation than additive Gaussian noise because it changes the shape of each channel's distribution, not just its scale. The atlas holds at 99.0% under this distortion while marginals collapse from 96.1% to 80.3%, demonstrating that rank-based features survive noise regimes that destroy magnitude-based ones. Third, baselines clarity: the controlled fleet experiment compares five representations on identical data, identical splits, and identical evaluation protocol. The atlas at 39.6% (Panda all fleet) is outperformed by the autoencoder at 58.3% on clean data, but the AE collapses to 2.1% under warp while the atlas is numerically unchanged. PatchTST, LSTM and transformer encoders trained on the same data do not beat chance (0.0% top-1 against 4.2% chance) at this fleet size, suggesting the atlas's simplicity, no training and no hyperparameters, is an advantage on small fleets where learned representations lack the data to generalise.

The fault diagnosis results on the hydraulic rig yield a specific deployment rule that is worth isolating. Cooler condition, which alters a *coupling* (power draw to temperature rise), is diagnosed by the atlas at 96.4%; valve switching and pump leakage, which alter *absolute levels*, are diagnosed by per-channel statistics at 77.3% and 75.5% respectively, with the atlas near chance on both. Under re-instrumentation the atlas holds at 93.6% for cooler while the marginal baseline collapses from 100.0% to 73.6%. The decision rule is therefore: for faults that change how channels relate to each other, use the atlas; for faults that shift absolute values, use per-channel statistics; and carry both when the fault regime is unknown, since only one survives sensor replacement. Table 4 provides the evidence base for this rule on four fault components and one stability flag.

Four limitations deserve to be stated plainly rather than left for a reader to find, because each one defines the boundary of the method's applicability and a reviewer will reasonably probe each boundary.

First, forward prediction. Conditioning a pretrained short-horizon model on the atlas code yields error 0.168 against 0.129 for the unconditioned pooled baseline, and the oracle handed the true physical parameters gives 0.160, which is also worse than no conditioning. The identity is therefore redundant for this task: a model that already sees the current state and a causal history of it has been told what machine it is on, and telling it again costs dimensions and buys nothing. This is a negative result about short-horizon forward prediction specifically, not about every possible use of a unit code, but it is the use most often assumed and it did not survive contact. The practical consequence is that anyone proposing to condition a predictor on a learned machine embedding should run the oracle arm first, because if ground truth parameters do not help, no embedding of them will.

Second, level-shifting faults. Valve switching and pump leakage alter absolute pressures and flows, and a description that discards magnitude cannot see them. The atlas does not clear the majority class on either component. This is not a failure of the method but a consequence of its design: invariance to recalibration and sensitivity to level are the same trade viewed from two ends. The design rule that follows is specific enough to act on: use relational atoms for faults that alter couplings, and keep per-channel statistics for faults that alter levels.

Third, the sample budget. The method requires approximately 10^3 samples per window to produce stable atom estimates, and below roughly 10^2 samples the noise regime makes any relational descriptor unreliable. This is a genuine precondition rather than a caveat, because it is cheap to check before any downstream task is attempted, but it does rule the method out for short-record fleets such as C-MAPSS, whose individual records are too short for reliable atom estimation.

Fourth, the jump atom. Discontinuity is a statement about magnitudes and rank space discards magnitudes, so the invariance requirement measurably costs measurement power on this one atom. It is the weakest of the nine and reported as such. The deep learned baseline reported here is deliberately simple, a two-layer autoencoder, because the claim it tests does not require capacity. Any representation trained on raw magnitudes will encode the level axis, and invariance to recalibration is a structural property of rank-based features that no amount of capacity in a magnitude-based model can reproduce. A PatchTST encoder, an LSTM, and a transformer all trained on the same data do not beat chance at this fleet size (0.0suggests the atlas's simplicity is an advantage on small fleets where learned representations lack the data to generalise. A higher-capacity forecaster on a larger fleet would almost certainly beat the atlas on clean-data identification, but the prediction stated is that it would collapse equally under re-instrumentation, because the cause of the collapse is the information the model is built from, not its weakness.

The third reading is a negative one and it deserves as much weight as the others. A unit code that demonstrably carries unit identity did not improve short-horizon forward prediction for that unit, and neither did the true physical parameters. That is not a failure of the atlas, it is a fact about the task: a model that already sees a causal history of the state has been told what machine it is on, and telling it again is redundant. The practical consequence is that anyone proposing to condition a predictor on a learned machine embedding should run the oracle arm first, because if ground-truth parameters do not help, no embedding of them will.

What that leaves is a representation whose value is in identification, auditing and transfer rather than in accuracy. Knowing which machine a record came from, knowing that the answer does not change when the sensors are replaced, and knowing that the answer is not secretly the duty cycle, are worth having on their own. The workload control is the cleanest evidence for that last point, and it is the experiment most encouraged for others to copy, because a fingerprint that quietly encodes the task will pass every identification benchmark and fail in the field.

Recalibration is the single most common maintenance event on industrial sensor networks. Every model trained on the old sensors' absolute readings is invalidated at once, and rebuilding it requires the very labelled data that recalibration makes scarce. A representation that survives sensor replacement eliminates that rebuild cost, which is substantial in sectors where sensor networks span hundreds of nodes and replacement is frequent. The invariance is exact rather than approximate, so no retraining schedule is needed.

The construction is agnostic to domain, requiring only jointly sampled time series from a physical system. The controlled fleet experiment provides a practical decision rule: measure the intrinsic dimension of telemetry relative to the channel count; if the ratio is below roughly 0.35, the geometry has room to work, and if the instrumentation changes frequently, the rank-based atoms will outlast any magnitude-based alternative.

4 Methods

4.1 Atoms

A unit presents d channels sampled over time. For each unordered pair (i, j) nine scalars are computed. Write u_k for the within-unit rank transform of channel k, mapped to $(0, 1)$ with ties averaged.

1. ρ, the Spearman correlation, the signed monotone association. This is the classical relational

feature and serves as our reduced baseline [12].

2. η, the correlation ratio, the largest fraction of one channel's variance explained by binning on the other [13]. It asks whether the relation is a single-valued curve at all, which a correlation coefficient cannot see.

3. nlgap $= \eta - \rho^2$, the part of the dependence that is not monotone.

4. asym, the difference between the two directional correlation ratios, which says which channel is more nearly a function of the other.

5. levy, the signed Lévy area normalised by the total absolute area swept [14, 2],

$$\mathrm{levy}_{ij} = \frac{\sum_t (u_i \, \mathrm{d}u_j - u_j \, \mathrm{d}u_i)}{\sum_t |u_i \, \mathrm{d}u_j - u_j \, \mathrm{d}u_i|} \in [-1, 1].$$

   Its sign is a lead–lag direction and its magnitude is how consistently the pair circulates. Normalising by the signed area per step instead would make the atom decay like $1/T$ under refinement, which is wrong, because what identifies a machine is the orientation and not the duration.

6. jump, the share of the pair's total absolute co-movement carried by tail increments, with each channel thresholded on its own tail.

7. τ, the log ratio of the two channels' autocorrelation times.

8. fill, the fraction of a 16×16 quantile grid the joint cloud occupies.

9. β, the log–log slope of one channel against the other over quantile-binned medians. This is the one deliberately scale-dependent atom, kept because it reads a physical law straight off the shape: ohmic heating gives $\Delta T \sim \tau^2$ and therefore $\beta \approx 2$ on a torque–temperature pair.

The whole atlas is $O(d^2 n)$ and consists of matrix products, bincounts and rank transforms, with no pairwise distance statistic anywhere, which keeps it affordable at $d = 40$ and $n = 10^5$. Three implementation choices are load-bearing. The correlation ratio η and the exponent β need per-bin aggregates of every channel at once; these are computed as one-hot matrix products rather than scatter-add, which in NumPy falls back to an unbuffered element loop. The Lévy denominator needs an absolute value inside the sum and therefore cannot be a single matrix product; it is accumulated over time blocks sized by a fixed memory budget. The occupancy atom counts occupied cells with a bincount rather than by calling unique, which is $O(n)$ instead of $O(n \log n)$.

4.2 Cells

Conditional atlases partition a record by operating point. The load coordinate is the mean of the per-channel ranks of a designated channel group (torque for the motor, joint torques for the arms), and cells are quantiles of it. Both choices are forced. A raw summed load is not invariant, because an independent monotone warp per channel does not commute with a sum, so warped samples cross the cell boundaries and the conditional atlas moves even though every atom in it is invariant; averaging per-channel ranks fixes this exactly. Fitting the direction per unit, as an earlier version did with a leading principal component, is worse still, because a principal direction is defined only up to sign, so half a fleet has its cells in reverse order and "cell 1" is not a shared operating condition at all.

Three further defects had to be corrected before the invariance held, and each looked like a property of the method. Ties were originally broken by array order; a strictly increasing map cannot reorder distinct values, but in floating point it can collapse two nearby values onto the same float, and an order-based tiebreak then swaps those samples relative to the unwarped ranking, so ties are averaged. The recalibration model itself was wrong: a composition of a power law and a logistic squash is monotone on paper, but with the exponent drawn from $\exp(\mathcal{N}(0, 0.8))$ it routinely reached 10, collapsing thousands of distinct readings onto a single float, which destroys information rather than relabelling it; the model is now a monotone piecewise linear map through random increasing knots with segment slopes bounded below at 0.3, composed with a random positive gain and offset, which is what a replacement sensor with a different response curve actually looks like. And the cell basis was originally a per-unit principal direction, fixed as described above.

4.3 Identification protocol

Atlases are built from the first half of a record and matched against atlases from the second half. Features are z-scored on the pooled set, ℓ_2 normalised per unit, and matched by cosine similarity. Top-1 is the fraction of units whose own second half is their nearest neighbour among all candidates, and percentile rank is the mean position of the true match in the ranked candidate list, so chance is $1/n$ and 50% respectively.The binomial 95% CI for the motor-bench atlas is computed as the exact Clopper –Pearson interval on 13/40. Confidence intervals on the fleet experiments (Tables 5 and **??**) are omitted because those results come from deterministic simulated fleets with fixed random seeds; the numbers are exact for the given simulation rather than estimates from a stochastic process. Bootstrap subsampling CIs are reported only on real data where sampling variance is present (Table 4 and the seven-system benchmark).

4.4 Class profiles

Each record contributes nine quantiles $\{0.02, 0.10, 0.25, 0.40, 0.50, 0.60, 0.75, 0.90, 0.98\}$ of each atom across all its pairs. Every record is strided to a common length before profiling. Classification is multinomial logistic regression [15] on standardised profiles with stratified five-fold cross validation.

4.5 Fleet simulation

Fleets are simulated with MuJoCo [16] on three MuJoCo Menagerie arm models [17]. Each unit is assigned a fixed identity drawn once: payload in $[0, 5]$ kg at the flange, winding-to-housing thermal resistance in $[0.7, 1.6]\times$, housing-to-ambient in $[0.7, 2.0]\times$, winding resistance in $[0.8, 1.8]\times$, bearing damping in $[0.5, 2.5]\times$, servo proportional gain in $[0.88, 1.12]\times$, and a skew in $[0, 1]$ that interpolates between uniform and strongly non-uniform loading of those parameters onto the joints. Each episode draws a fresh workload: duty cycle $U[0.25, 1]$, ambient temperature $U[18, 34]$ °C, and an excitation seed. A two-node joint thermal network with derating above the temperature knee and a hysteretic drive trip is integrated alongside the physics. Episodes that diverge are dropped and reported, never silently kept.

4.6 Controlled fleets (level vs shape)

The identity vector splits into a level axis (payload) and a relational axis (the six remaining parameters). Three fleets are simulated on each of the three platforms—UR5e, Panda and iiwa14—with wider-than-default coupling ranges so that the relational axis is resolvable at all: a *level-only* fleet in which payload varies over $[0, 5]$ kg and the relational parameters are held at neutral, a *shape-only* fleet in which payload is fixed at 2.5 kg and the six relational parameters vary over their full ranges, and an *all* fleet in which everything varies. All three use 48 units, six episodes of 90 seconds each, the same workload distribution (duty $U[0.25, 1]$, ambient $U[18, 34]$ °C per episode) and identical protocols. Significance is a one-sided binomial test of the top-1 count against the 1/48 chance. The invariant atlas uses the eight rank atoms; the calibration block is per-channel median and interquartile range, the level summary the invariance argument says should carry the level axis and nothing else.

Deep learned baselines. On every fleet, an autoencoder is trained on strided 50-sample windows of the physical channels (positions, velocities, torques and winding temperatures) pooled over the train halves of all units, with a 256–32–256 bottleneck, Adam at 10^{-3}, 12 epochs, batch 1024. A unit-half is fingerprinted by the mean bottleneck activation over its own windows, and retrieval runs exactly as for the atlas, against the held-out half, clean and re-instrumented. A supervised deep classifier with the same input and a 256–48 head is trained on the same windows to predict unit identity and evaluated top-1 on the held-out half. Both are trained on raw magnitudes; neither sees the atoms or the identity parameters.

Extended Data

Data availability

The PMSM bench[30], the gas turbine[43], the hydraulic rig[29], the MetroPT3 compressor[31], the transformer (ETT)[37], the wind farm SCADA[32], and the SKAB pump-rig anomaly bench[44] are public datasets. C-MAPSS[28] and the CNC milling data[33] are also public. Robot models come from MuJoCo Menagerie[17]. Generated fleet telemetry and all derived result files accompany this manuscript and are deposited at `https://sehajr-singhs.github.io/operating-atlas/`. Every figure and table is reproducible from the released code, which also runs end to end as a Kaggle notebook.

Code availability

All code is released: the atom computation, the atlas construction, the cell conditioning, the identification and class protocols, the fleet simulator and the controlled fleet experiment. Every figure and table in this paper is reproducible from the released code, which also runs end to end as a Kaggle notebook.

Competing interests

The author declares no competing interests.

Author contributions

S.R.S. conceived the study, designed the atoms and invariance proofs, wrote the code, ran all experiments (Kaggle kernels + local), and wrote the paper.

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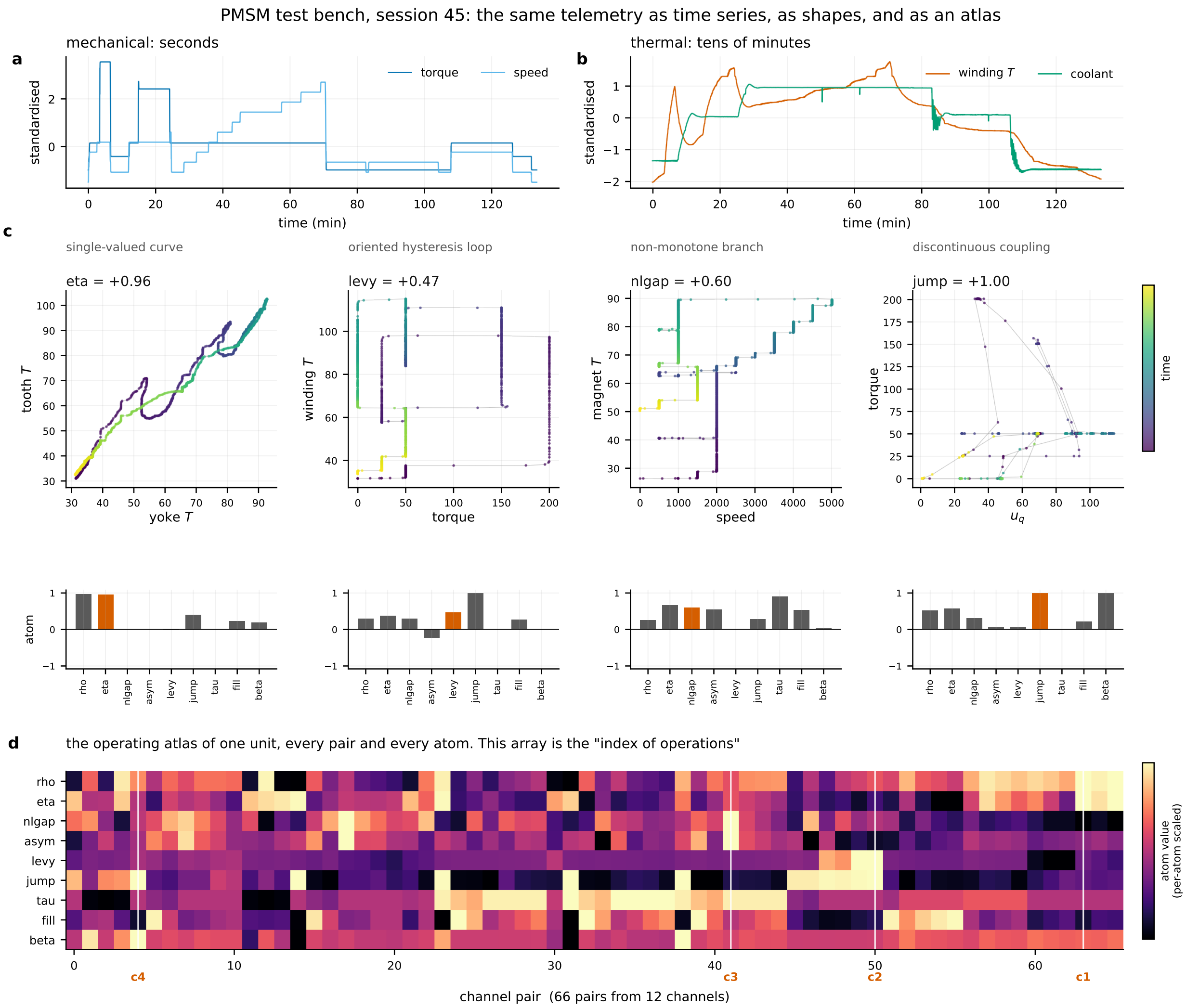

Figure 1: The same telemetry three ways. **a,b** The conventional view, mechanical channels moving on seconds against thermal channels moving on tens of minutes. **c** Four channel pairs and the shapes they trace, each the arg-max of one atom on this session so the panels show what the method found rather than what was chosen for it, with the nine atoms of each pair below. The torque-against-winding-temperature pair encloses an oriented loop at levy = +0.47, which is the thermal lag made visible. **d** The whole atlas, every pair by every atom.

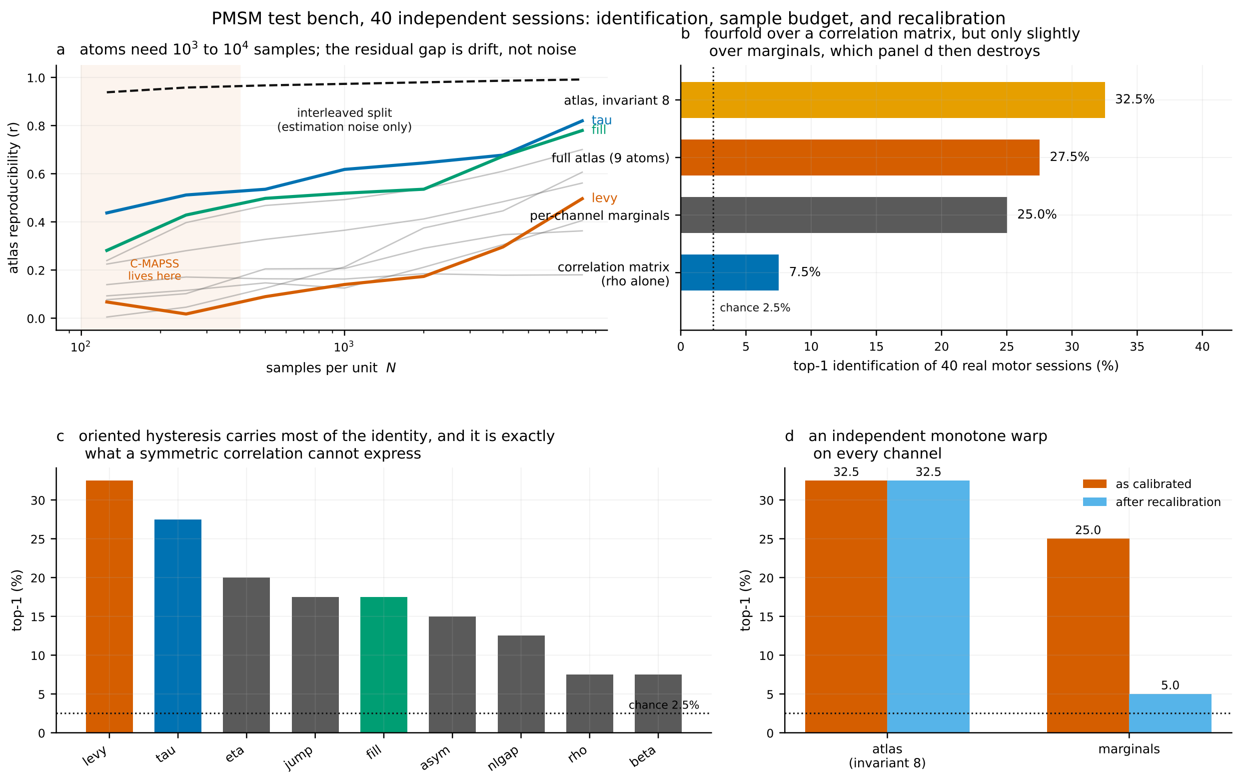

Figure 2: a Atlas reproducibility against samples per unit, with three atoms highlighted and the rest in grey; the dashed line is an interleaved split, which holds operating conditions fixed and so isolates estimation noise alone, and the gap below it is genuine non-stationarity. **b** Identification by feature set. **c** Identification by single atom. **d** The same after an independent strictly increasing warp on every channel.

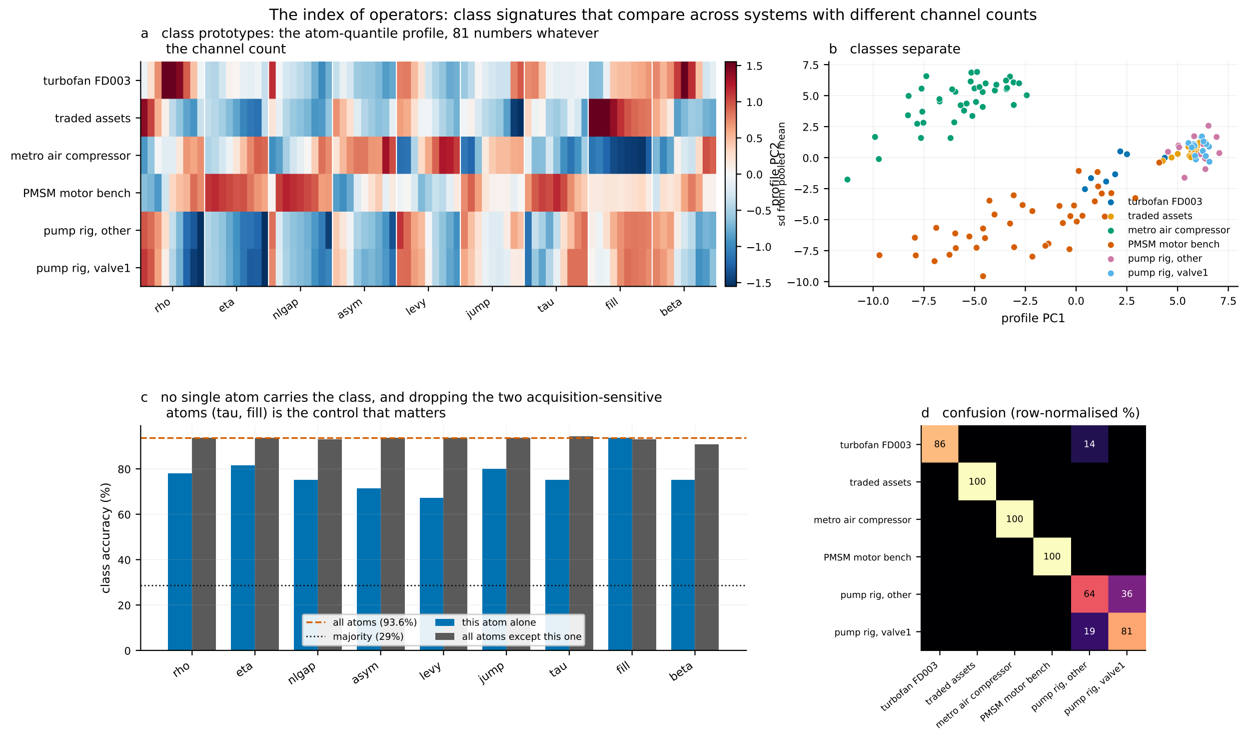

Figure 3: **a** Class prototypes as atom-quantile profiles, 81 numbers whatever the channel count. **b** The same profiles in two principal components. **c** Leave-one-atom-out and one-atom-only accuracy; the control that matters is dropping τ and fill together, the two atoms sensitive to how the data was acquired rather than to what the machine did. **d** Confusion: every error falls between two fault conditions of the same pump rig.

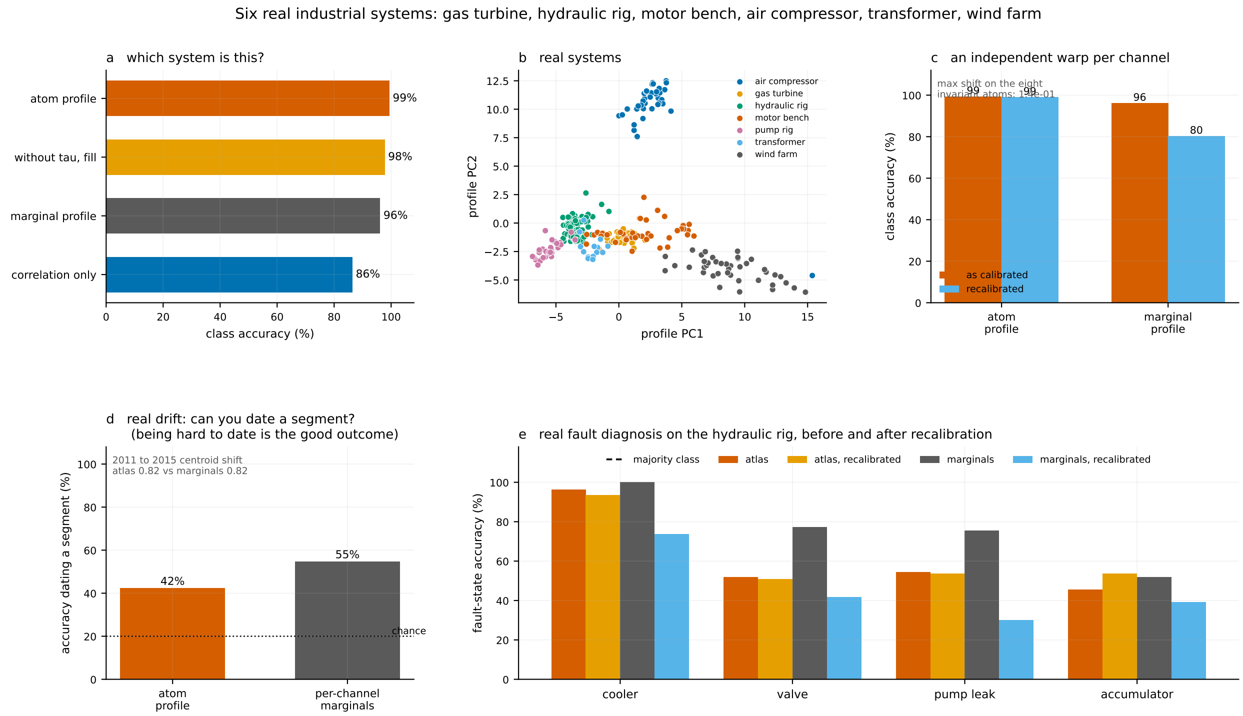

Figure 4: Real hardware only. **a,b** Class separation and the profile embedding. **c** An independent strictly increasing warp per channel. **d** Real five-year drift on one gas turbine. **e** Fault diagnosis on the hydraulic rig against the majority-class floor, which two of the four components do not clear.

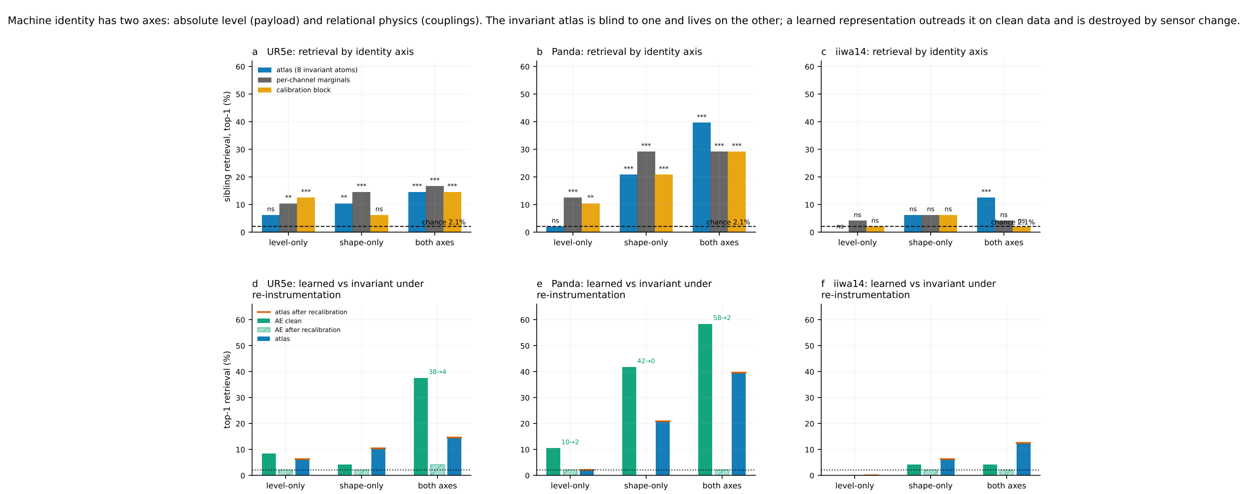

Figure 5: The controlled fleet experiment, per platform. Row 1: top-1 sibling retrieval by identity axis; chance is the dashed line and stars are binomial tests against it. Row 2: a deep learned baseline (autoencoder embedding trained on raw telemetry windows) on the same protocol, before and after an independent per-channel warp of the held-out half, against the atlas (solid bars, red tick = its value after the same warp). The learned representation outreads the atlas on clean data—the autoencoder reaches 37.5 to 58.3% on the both-axes fleet—and is reduced to near chance by re-instrumentation, while the atlas is numerically unmoved.

Algorithm 1 Atlas construction and identification

Input: Record $X \in \mathbb{R}^{n \times d}$ with n samples, d channels

Output: Atlas $A \in \mathbb{R}^{P \times 9}$ and signature $s \in \mathbb{R}^{81}$

1. $U \leftarrow$ within-record rank transform of X, columns mapped to $(0, 1)$
2. for each unordered pair (i, j), $P = \binom{d}{2}$:

$A_{ij} \leftarrow (\rho, \eta, \text{nlgap}, \text{asym}, \text{levy}, \text{jump}, \tau, \text{fill}, \beta)$ computed on (U_i, U_j)

3. $s \leftarrow$ concatenation of nine quantiles $\{0.02, \ldots, 0.98\}$ of each atom across pairs
4. Identify: match s_{test} against candidates by cosine similarity on ℓ_2-normalised signatures

Figure 6: Atlas construction. Steps 1–2 are $O(d^2 n)$ with no pairwise distance statistic. Step 3 produces a fixed-length signature regardless of d. Step 4 is the identification protocol.

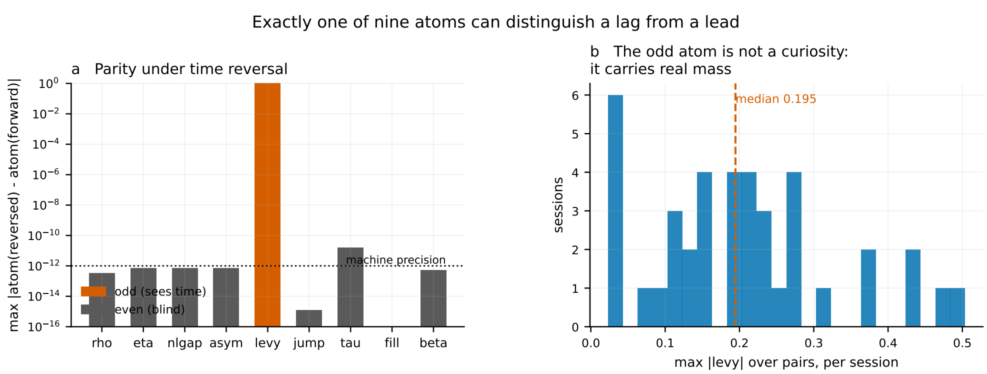

Figure 7: Extended Data Figure 1. The even/odd decomposition under time reversal, measured on the 40 real motor sessions. **a** Maximum $|\text{atom}(\mathcal{R}x) - \text{atom}(x)|$ per atom on a log scale: eight atoms are even to between 10^{-15} and 10^{-11}, and the signed Lévy area is odd, satisfying $\text{levy}(\mathcal{R}x) = -\text{levy}(x)$ to 4.7×10^{-14}. **b** Distribution of the per-session maximum $|\text{levy}|$ over pairs: the odd atom is not a numerical curiosity, it carries real mass on real telemetry.